

Assignment –4

Unit-1 CLT

1. The lifetime of a transistor is a random variate with mean 2500 hours and standard deviation 300 hours. Using CLT find the probability that the average life time of 50 transistors is greater than 2550 hours. Ans :0.1190
2. Let $X_1, X_2, \dots, X_{100}$ be 100 independent and identically distributed(iid) random variables with mean 2 and variance $\frac{1}{4}$. find $P(190 < X_1 + X_2 + \dots + X_{100} < 210)$. Ans: 0.9544
3. The lifetime of a certain make of lamp is distributed exponentially with mean of 30 hours. Find the probability that 144 of these lamps will give total light for more than 4500 hours. Ans: 0.3085
4. If $X_1, X_2, \dots, X_n$ are poisson variates with the parameter $\lambda = 2$, use CLT to estimate $P(120 < S_n < 160)$ where $S_n = X_1 + X_2 + \dots + X_n$ and $n=75$. Ans:0.7868
5. A bank teller serves customers standing in the queue one by one. Suppose that the service time X_i for customer i has mean $EX_i=2$ (minutes) and $Var(X_i)=1$. We assume that service times for different bank customers are independent. Let Y be the total time the bank teller spends serving 50 customers. Find $P(90 < Y < 110)$. Ans: 0.8427
6. A population of 29 year-old males has a mean salary of \$29,321 with a standard deviation of \$2,120. If a sample of 100 men is taken, what is the probability their mean salaries will be less than \$29,000? Ans:0.07